

How Graphene Is CHANGING INDUSTRIES

Graphene is reshaping industries—extending the electric vehicle range, cutting data centre energy costs, and sparking quantum tech innovations. This article is based on EFY’s Ashwini Kumar Sinha and Nidhi Agarwal’s conversation with John Tingay of Paragraf to discuss the science driving this shift.



JOHN TINGAY, Paragraf’s CTO, leads research and development with expertise in open innovation management, the transfer and setup of SMEs, and technical leadership in defining complex system architectures.



A technician at a Paragraf lab

Graphene is a material poised to revolutionise the electronics and semiconductor industries. Once a laboratory curiosity, it now faces the critical challenge of scaling production—from high-performance prototypes to millions of reliable devices. This is not just about numbers; maintaining quality on a mass scale is key to graphene’s economic viability.

One early breakthrough in scalable graphene technology is a magnetic sensor designed for cryogenic environments and mainstream applications like electric vehicles (EVs). The same material that could one day enhance your smartphone is already improving EV sensors, offering

superior sensitivity, low noise, and a broad dynamic range—perfect for the market’s shift away from internal combustion engines.

But let us not stop there. What about graphene’s potential beyond electronics? The material is also making strides in bio-sensing and molecular applications, areas rich with academic research now transitioning into practical, scalable platforms. This move could open up new markets and applications that were previously unimaginable. In a conversation with John Tingay, the CTO of Paragraf, we explored the possibilities and envisioned a future where the full potential of graphene is realised.

When discussing graphene production, John says, “We use metal organic chemical